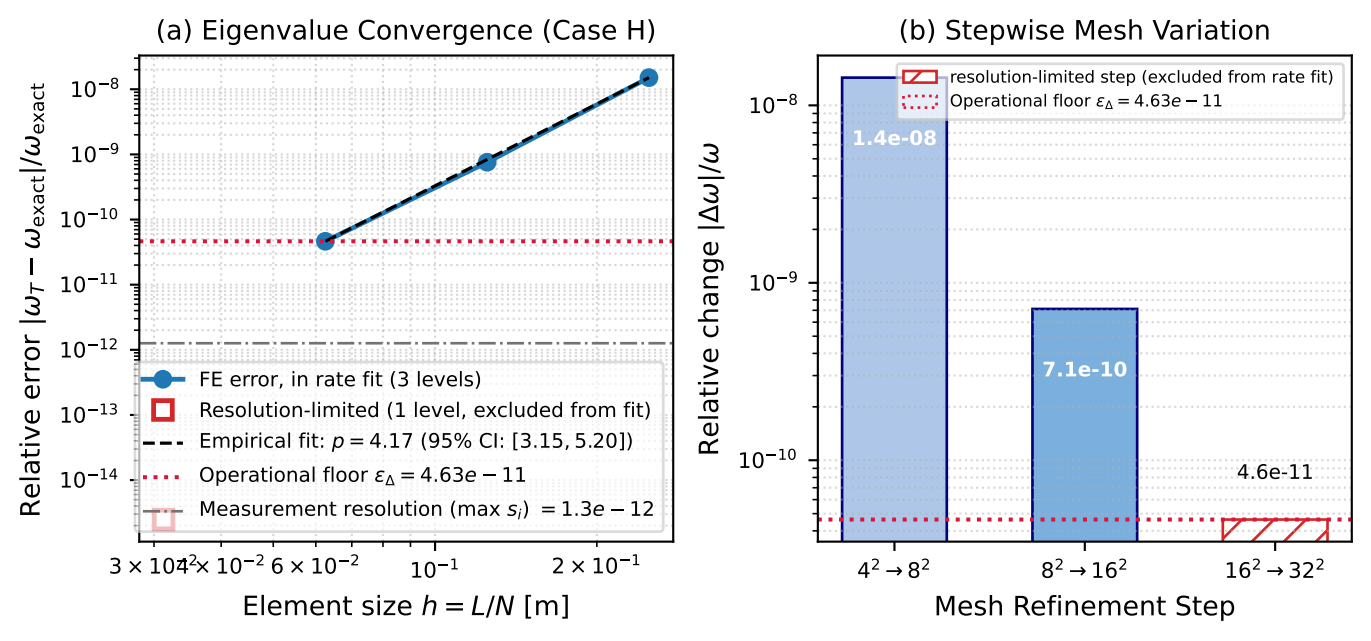
(a) Eigenvalue Convergence (Case H)
Relative error $|\omega_T - \omega_{\text{exact}}|/\omega_{\text{exact}}$
10^{-8}
10^{-9}
10^{-10}
10^{-11}
10^{-12}
10^{-13}
10^{-14}
FE error, in rate fit (3 levels)
Resolution-limited (1 level, excluded from fit)
Empirical fit: $p = 4.17$ (95% CI: [3.15, 5.20])
Operational floor $\varepsilon_\Delta = 4.63e-11$
Measurement resolution (max s_i) $= 1.3e-12$
3×10^{-2}
4×10^{-2}
6×10^{-2}
10^{-1}
2×10^{-1}
Element size $h = L/N$ [m]
(b) Stepwise Mesh Variation
Relative change $|\Delta\omega|/\omega$
resolution-limited step (excluded from rate fit)
Operational floor $\varepsilon_\Delta = 4.63e-11$
10^{-8}
10^{-9}
10^{-10}
1.4e-08
7.1e-10
4.6e-11
$4^2 \to 8^2$
$8^2 \to 16^2$
$16^2 \to 32^2$
Mesh Refinement Step